

Reducing Hallucinations via Citation-Grounded QA with Locally-Hosted Small Language Models

CS505 Natural Language Processing | Final Project Proposal | Boston University

[We will add names here :)]

Project type: Empirical NLP project focused on retrieval-augmented generation (RAG) for hallucination reduction.

Research question: *How much of the hallucination gap between a locally-hosted small language model (SLM) and GPT-4o is attributable to retrieval quality versus generation capacity—and can systematic retrieval improvements make fully local, privacy-preserving deployment practically viable?*

1 Problem Statement & Motivation

Hallucination—generating claims unsupported by source documents—is the primary failure mode of LLMs in high-stakes settings. Cloud tools like Google’s NotebookLM answer questions over private documents with inline citations, but require transmitting the entire corpus to an external server, violating HIPAA, GDPR, and corporate data-governance policies. **Cloud upload equals data-exfiltration risk.** Sub-7B SLMs (e.g., Phi-3-mini-3.8B [Microsoft, 2024]) can run on a single consumer GPU, but it is unclear whether their higher hallucination rate stems from *retrieval failures* or *generation failures*. If retrieval improvements close most of the gap, local deployment becomes a practical alternative.

Task definition. Given a corpus $\mathcal{C} = \{d_1, \dots, d_n\}$ and query q , the system produces answer a plus, for every sentence $s_i \in a$, a citation (p_i, s_i) where passage $p_i \in \mathcal{C}$ textually entails s_i . Any sentence without a valid citation is counted as a hallucination.

2 Proposed System

All components run locally (no API calls except the GPT-4o oracle).

Parsing. PyMuPDF; preserves section headings for structure-aware chunking.

Chunking (ablated). Three strategies: fixed-size (256 tok, 32 overlap), sliding-window (512 tok, 128 overlap), section-aware (boundary-aligned).

Retrieval (ablated). Dense-only (FAISS + BGE-small-en-v1.5); BM25 (rank-bm25); hybrid reciprocal-rank fusion (RRF); hybrid + cross-encoder reranking (ms-marco-MiniLM-L-6-v2), top-5 passages.

Generation. Two prompt variants: standard generation (no citation instruction) vs. *citation-forcing* (the model must bracket every claim with an inline citation and assert only what the retrieved passages support). Served via Ollama; ≤ 16 GB VRAM. If Phi-3-mini fails to follow the bracket format consistently, we will fall back to constrained decoding (Guidance or Outlines) to enforce the schema at the token level.

3 Experimental Design

3.1 Conditions

Each condition changes exactly one component (Table 1). Conditions A–E are the committed core; F and G are stretch goals.

3.2 Dataset & Baselines

Dataset: QASPER [Dasigi et al., 2021]—1,585 NLP papers, 5,049 Q&A pairs with gold evidence annotations supporting both retrieval and answer evaluation (official token F1).

Baselines: (1) **Parametric-only LM**—Phi-3-mini with no retrieval, establishing the hallucination floor.

Cond.	Chunk	Retrieval	Rerank	Cite-force	Generator
A (base)	Fixed	Dense	×	×	Phi-3-mini
B	Fixed	Hybrid	×	×	Phi-3-mini
C	Section	Hybrid	×	×	Phi-3-mini
D	Section	Hybrid	✓	×	Phi-3-mini
E (best)	Section	Hybrid	✓	✓	Phi-3-mini
F*	Section	Hybrid	✓	✓	Llama-3-8B
G*	Section	Hybrid	✓	✓	GPT-4o (oracle)

Table 1: Ablation conditions. *Stretch goals (time permitting).

(2) **BM25-only**—sparse keyword retrieval, the simplest retrieval baseline. (3) **Random Retrieval**—five passages sampled uniformly at random, confirming that RRF gains reflect retrieval *quality* rather than the mere presence of context.

3.3 Metrics & Hypothesis

Retrieval: Recall@5, MRR@10, evidence hit rate. Answer: Token F1 (QASPER official). Faithfulness: RAGAS [Es et al., 2023]. Citation precision: NLI entailment (TRUE cross-encoder), supplemented by a lightweight human evaluation on 20–30 sampled QA pairs to guard against NLI model biases.

Hypothesis: Retrieval improvements (A→D) will reduce hallucination more than citation-forcing alone (D→E), identifying retrieval as the primary bottleneck; section-aware chunking will boost citation precision.

4 Related Work

Lewis et al. [2020] introduced RAG and remains the canonical reference for retrieval-augmented generation; Asai et al. [2024] is the closest prior system, applying RAG to scientific literature synthesis. Our contribution—a controlled ablation on a locally-hosted SLM that disentangles retrieval failures from generation failures—is absent from both works and from the broader attribution evaluation literature.

5 Feasibility & Timeline

Scope. The committed core (A–E on QASPER with Phi-3-mini) builds the retrieval pipeline incrementally—one component per week—and is realistic on a single consumer GPU. Stretch goals (F, G, embedding ablation, human eval) will be attempted in week 5 and deferred to the final report if time is short.

Wk	Milestone
1–2	Parsing, chunking strategies, BM25 + dense baseline (cond. A)
3	Hybrid RRF + reranking; conditions B–D; QASPER eval pipeline
4	Citation-forcing (E); RAGAS + NLI evaluation; error analysis
5	Midway report; begin stretch goals (F, G) if on schedule
6	Final ablation, writeup (ACL format)

Table 2: Six-week schedule. Core scope: weeks 1–4 (conditions A–E).

Stack: PyMuPDF, sentence-transformers, FAISS, rank-bm25, Ollama, RAGAS, HuggingFace Transformers; no paid API for core scope.

References

- A. Asai et al. OpenScholar: Synthesizing scientific literature with retrieval-augmented language models. *arXiv preprint arXiv:2411.14199*, 2024.
- P. Dasigi, K. Lo, I. Beltagy, A. Cohan, N. A. Smith, and M. Gardner. A dataset of information-seeking questions and answers anchored in research papers. In *Proceedings of NAACL*, 2021.

- S. Es, J. James, L. Espinosa-Anke, and S. Schockaert. RAGAS: Automated evaluation of retrieval augmented generation. *arXiv preprint arXiv:2309.15217*, 2023.
- P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. arXiv:2005.11401.
- Microsoft. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024.